

Underneath the Black Marker

AI Techniques for Detecting and Reversing Redactions

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The Usual Disclaimer...

My thoughts are my own! This session is presented for informational and educational purposes only. It may include content or use devices that serve merely to stimulate discussion and thought such as hyperbole, understatement, humor, devil's advocacy, figurative language, and the like. The views, thoughts, opinions, information, and materials comprising the presentation do not necessarily represent the opinions of each presenter's employer, partners, clients, customers, or membership. The content does not set policies, establish standards of care, or create guidelines. Nothing in this session should be construed as legal advice or an offer of representation. The presentation is not a substitute for independent professional judgment or legal advice concerning the law of a particular jurisdiction as applied to the full factual context of a unique situation or matter.

Agenda

- **Why & The Downright Laughable** 
 - **Demos! Demos! Demos!**
- **Pulling on Loose Threads**
 - **Demos! Demos! Demos!**
- **Reversing Obfuscation on Images**
 - **Demos! Demos! Demos!**
- **Pushing the Envelope Using AI/ML**
 - **Demos! Demos! Demos!**
- **Q & A**

Why I Decided To Do This Talk...



24/7 Live | New York City | New Jersey | Long Island | Northern Suburbs | Connecticut

Q | Bell | User

JEFFREY EPSTEIN

Justice Department says it's taken down Epstein-related files that may have had victim information

The lawyers represent more than 200 alleged Epstein victims.



Jeffrey Epstein



Judge to weigh Bank of America settlement with Epstein accusers



Judge rejects Ghislaine Maxwell's attempt to overturn conviction



New Mexico attorney general sues DOJ over Epstein files he says are needed for state...



Model recruiter who introduced women to Jeffrey Epstein found dead

Watch Live

Image Attribution

<https://abc7ny.com/post/epstein-victims-lawyers-ask-court-order-doj-take-down-files-website/18527173/>

...But Protecting Rich and Powerful

@davidvonthenen



Image Attribution

<https://www.pbs.org/newshour/politics/the-facts-and-timeline-of-trump-and-epsteins-falling-out>

Downright Laughable 🤣

The Initial Release...

Image Attribution
<https://www.cnn.com/politics/live-news/jeffrey-epstein-files-released>



CNN Politics

Elections 2026

Trump

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December 19, 2025 — Jeffrey Epstein files released

CNN reporters review the trove that stems from the investigation into the convicted sex offender

Updated 1:15 AM EST, Sat December 20, 2025



Recalled Dec 20–21, 2025

Image Attribution

<https://www.pbs.org/newshour/politics/at-least-16-files-disappear-from-doj-site-for-epstein-documents-including-trump-photo>

Note on
(e.g., handwritten
may produce unreliable

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By —

Michael R.
Sisak,
Associated
Press

By —

David B.
Caruso,
Associated
Press

At least 16 files disappear from DOJ site for Epstein documents, including Trump photo

Politics Dec 20, 2025 6:55 PM EDT

NEW YORK (AP) — At least 16 files disappeared from the Justice Department's public

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feedback

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Second Batch Jan 30, 2026

Image Attribution
<https://www.nbcnews.com/politics/justice-department/doj-releases-new-trove-long-awaited-epstein-files-rcna256714>

[POLITICS](#)[U.S. NEWS](#)[WORLD](#)[LOCAL](#)[SPORTS](#)[BUSINESS](#)[HEALTH](#)[TIPLINE](#)[SHOPPING](#)[WATCH](#)[SUBSCRIBE](#)[Epstein files](#)[Live updates](#)[Millions of pages released](#)[What we already knew](#)[Epstein estate photos](#)

EPSTEIN FILES

DOJ releases millions of pages of additional Epstein files

The latest disclosures come more than a month after the Justice Department posted 3,500 files on its website.



Listen to this article with a [free profile](#)

00:00

08:12



1x



3 million pages of Epstein documents released

02:31



Recall Again: Feb 6, 2026

Image Attribution

<https://www.businessinsider.com/doj-jeffrey-epstein-removed-thousands-files-2026-2>
<https://www.bbc.com/news/articles/cn0k65pnxix0>

BUSINESS INSIDER

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Networks Boycott Trump



H-1B Fallout: 1 Year Later

Squishy Nightmare

Small Business Insider

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The Long Play

NEW

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NEW

LAW

The DOJ says it took down over 9,500 of the Epstein files

By [Aditi Bharade](#) + and [Jacob Shamsian](#) +

BBC

Home News Sport Business Technology Health Culture Arts Travel Earth Audio Video Live Docu

Thousands of Epstein documents taken down after victims identified

4 February 2026

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Jessica Rawnsley



Demo: Copy and Paste

https://youtu.be/_xf3SEzj4ow

Demo: Programmatic Access

<https://youtu.be/mviXN9DYSZA>

Pulling on Loose Threads

What Family Think Hacking Is...

Video Info
[https://en.wikipedia.org/wiki/Hackers_\(film\)](https://en.wikipedia.org/wiki/Hackers_(film))

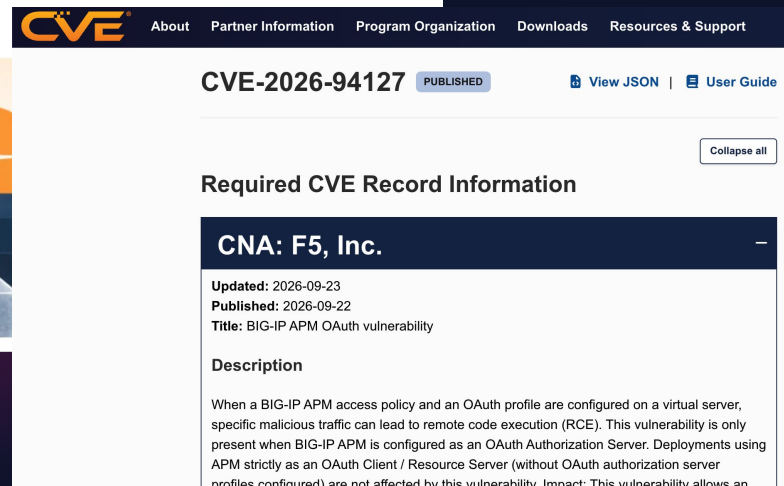
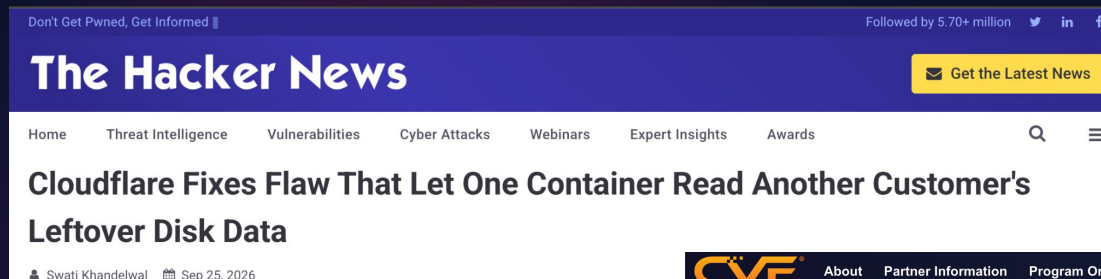


The Reality of Hacking...

Video Info
[https://en.wikipedia.org/wiki/Sneakers_\(1992_film\)](https://en.wikipedia.org/wiki/Sneakers_(1992_film))



Recent Examples of Real Attacks



<https://thehackernews.com/2026/09/cloudflare-fixes-flaw-that-let-one.html>

<https://arstechnica.com/security/2026/03/widely-used-trivy-scanner-compromised-in-ongoing-supply-chain-attack/>

<https://labs.watchtower.com/is-this-a-joke-in-the-auth-header-f5-big-ip-unauth-heap-overflow-to-rce-cve-2026-94127/>

Which Threads To Pull On?

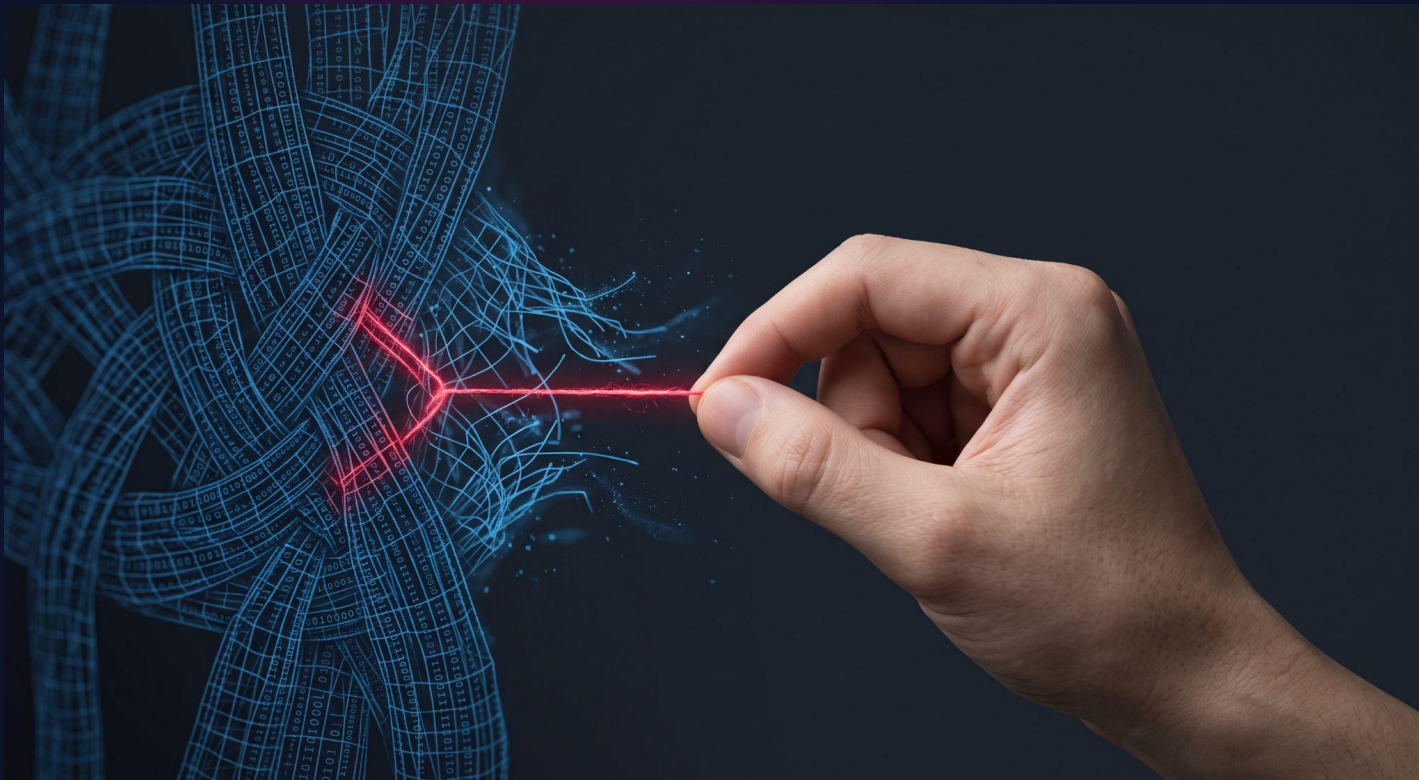


Image Attribution
Gemini Generated

Majority of Files are PDFs

```
(kali@kali)-[~]
$ ls -l Downloads/Optiv_PoC_Dompdf.pdf
-rw-r--r-- 1 kali kali 1468 Aug  2 17:30 Downloads/Optiv_PoC_Dompdf.pdf

(kali@kali)-[~]
$ exiftool Downloads/Optiv_PoC_Dompdf.pdf
ExifTool Version Number      : 12.41
File Name                    : Optiv_PoC_Dompdf.pdf
Directory                    : Downloads
File Size                    : 1468 bytes
File Modification Date/Time  : 2022:08:02 17:30:20+05:30
File Access Date/Time       : 2022:08:02 17:30:45+05:30
File Inode Change Date/Time  : 2022:08:02 17:30:45+05:30
File Permissions             : -rw-r--r--
File Type                    : PDF
File Type Extension          : pdf
MIME Type                    : application/pdf
PDF Version                  : 1.7
Linearized                   : No
Page Count                   : 1
Producer                     : dompdf 1.2.0 + CPDF
Create Date                  : 2022:08:02 11:57:58+00:00
Modify Date                  : 2022:08:02 11:57:58+00:00

(kali@kali)-[~]
$
```

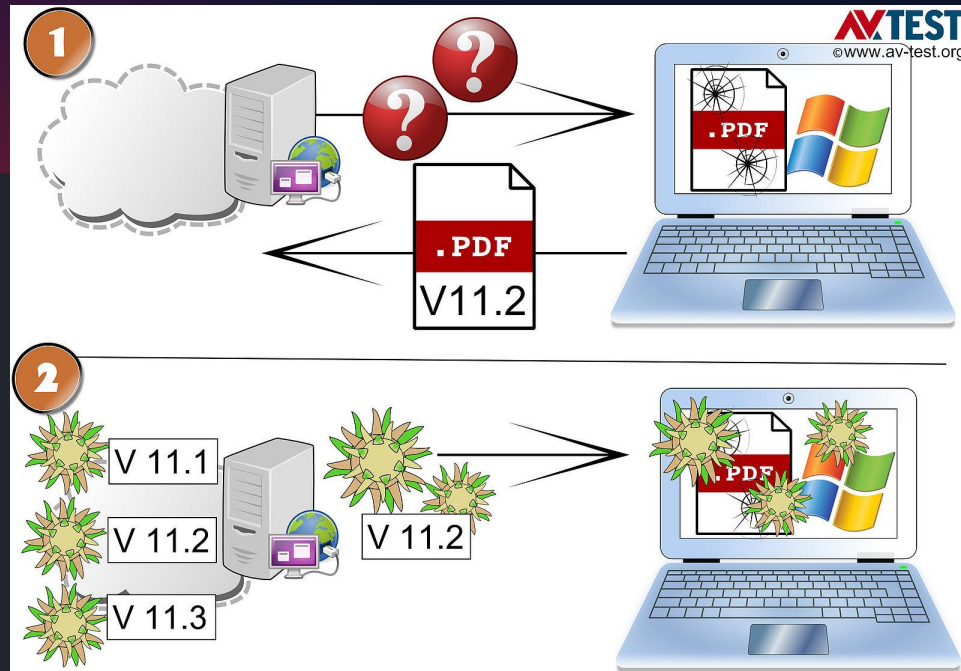


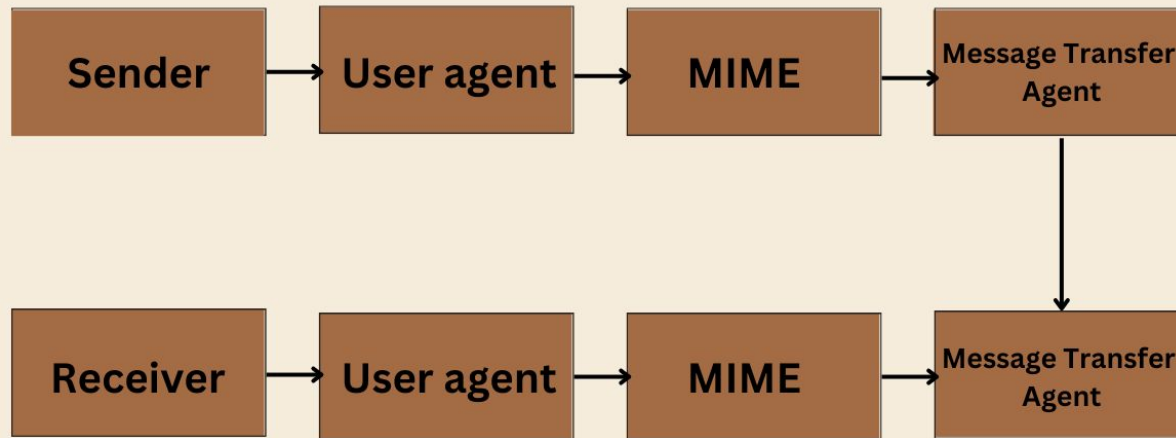
Image Attribution

<https://www.av-test.org/en/news/adobe-java-make-windows-insecure/>
<https://www.optiv.com/insights/discover/blog/exploiting-rce-vulnerability-dompdf>

Most Communications Via Email



MIME PROTOCOL



Multipart/Mixed

Headers

From:
To:
Subject:
Date:

Message-id;
Mime-version;
Content-type;

Parts

Text Parts

Text/Plain

Content

Text/HTML

Content

Attachments

Attach. 1

Content

Attach. N

Content

Image Attribution

<https://www.naukri.com/code360/library/mime-protocol>

<https://docs.aspose.com/email/net/email-file-formats/>

Demo: Exploiting PDF Versioning

<https://youtu.be/v5uDC6x1EBU>

Demo: Exploiting PDF Metadata

<https://youtu.be/EZEejTQkO2U>

Demo: Email Protocols

<https://youtu.be/YCjzrl-J4P0>

Idea: Data Locality & Semantic + NER

What is Data Locality on Physical Disk?

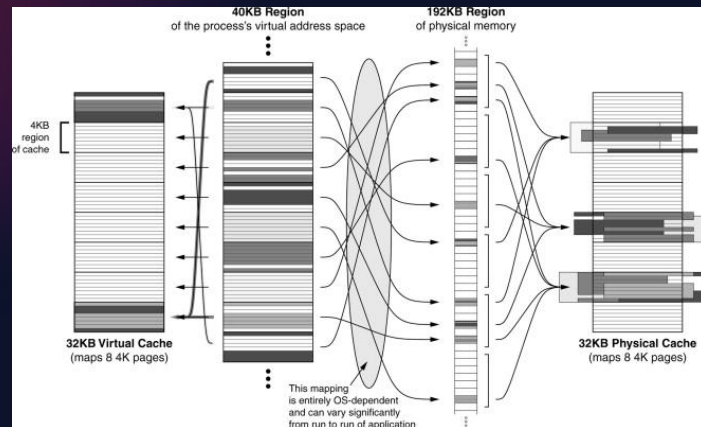
- Storing Related Data Blocks Close to Each Other

This Mimics the Real World Because:

- Typically Have Clusters of Communication
- Ex: First Text Message → Hundreds → Drops Off

Bet This Works For Email Too

- Names Correlated By Locality
- Fit Names to Black Box Sizes



Reversing Obfuscation on Images

What About Images?

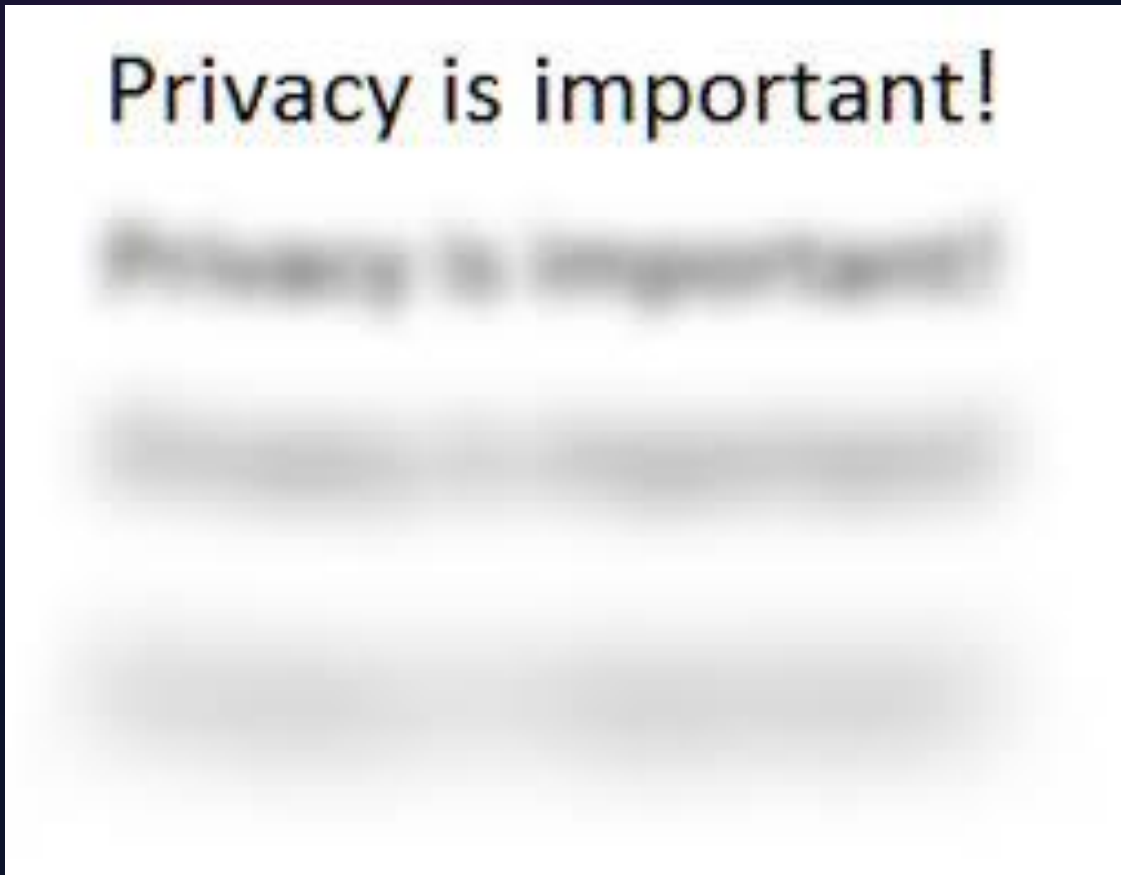


Image Attribution

https://www.researchgate.net/figure/Example-of-Gaussian-blurred-text-with-blur-radii-from-top-to-bottom-of-0p-15p-30p_fig5_305423573

Look at Depixelization...

https://codeberg.org/spipm/Depixelization_poc

- Defined Grid of Pixels, Averages the Color Value
- Reference Fonts Are Consistent, Same Pixelated Box

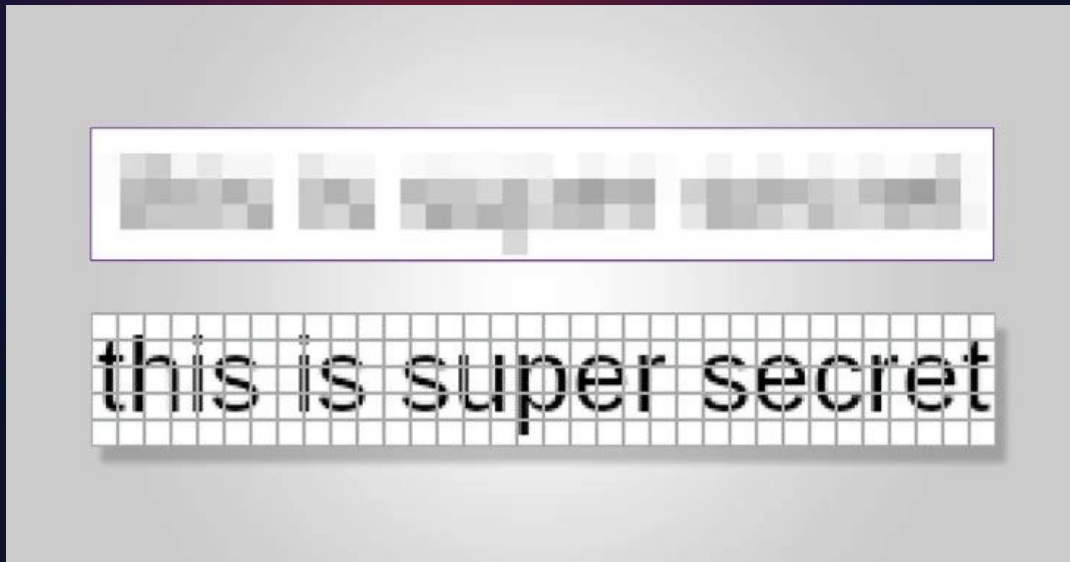


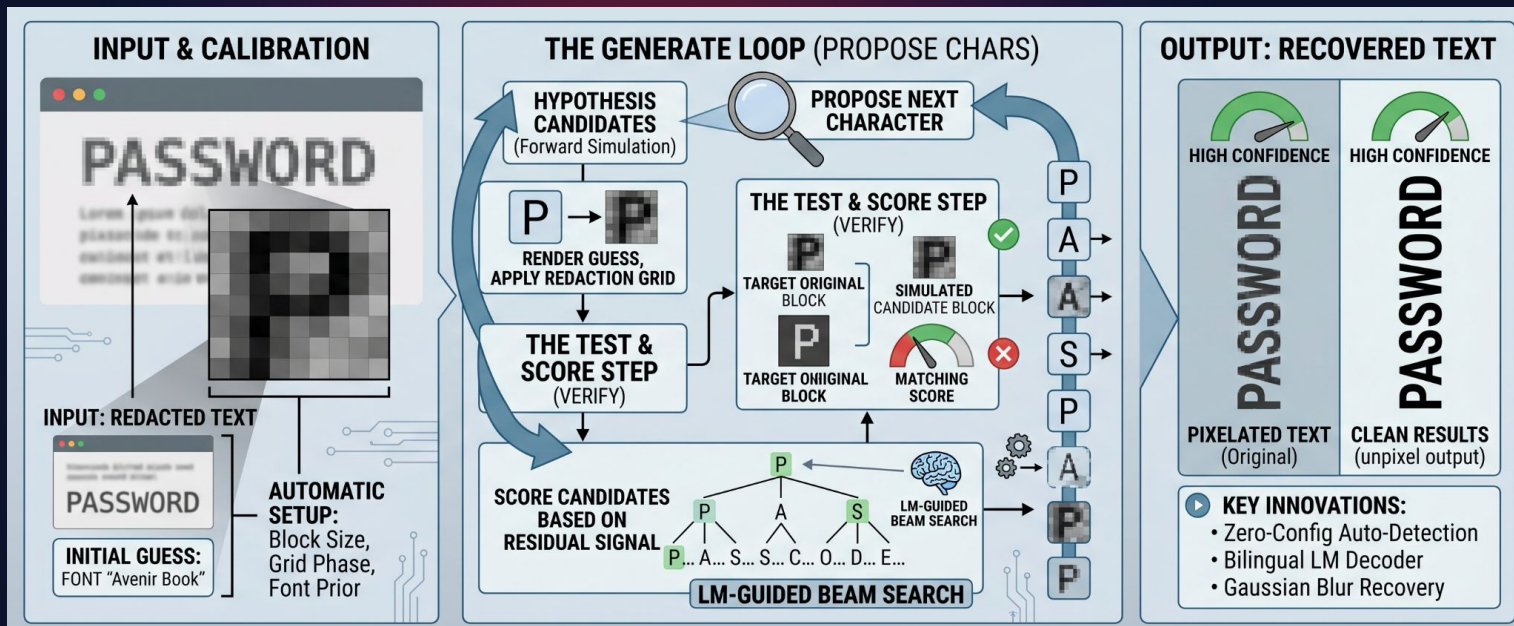
Image Attribution

<https://thehackernews.com/2022/02/this-new-tool-can-retrieve-pixelated.html>

...and Unpixel

<https://github.com/oioio-space/unpixel>

- Uses a Generate-and-Test Approach (Render→Match)



Or Video?



Image Attribution

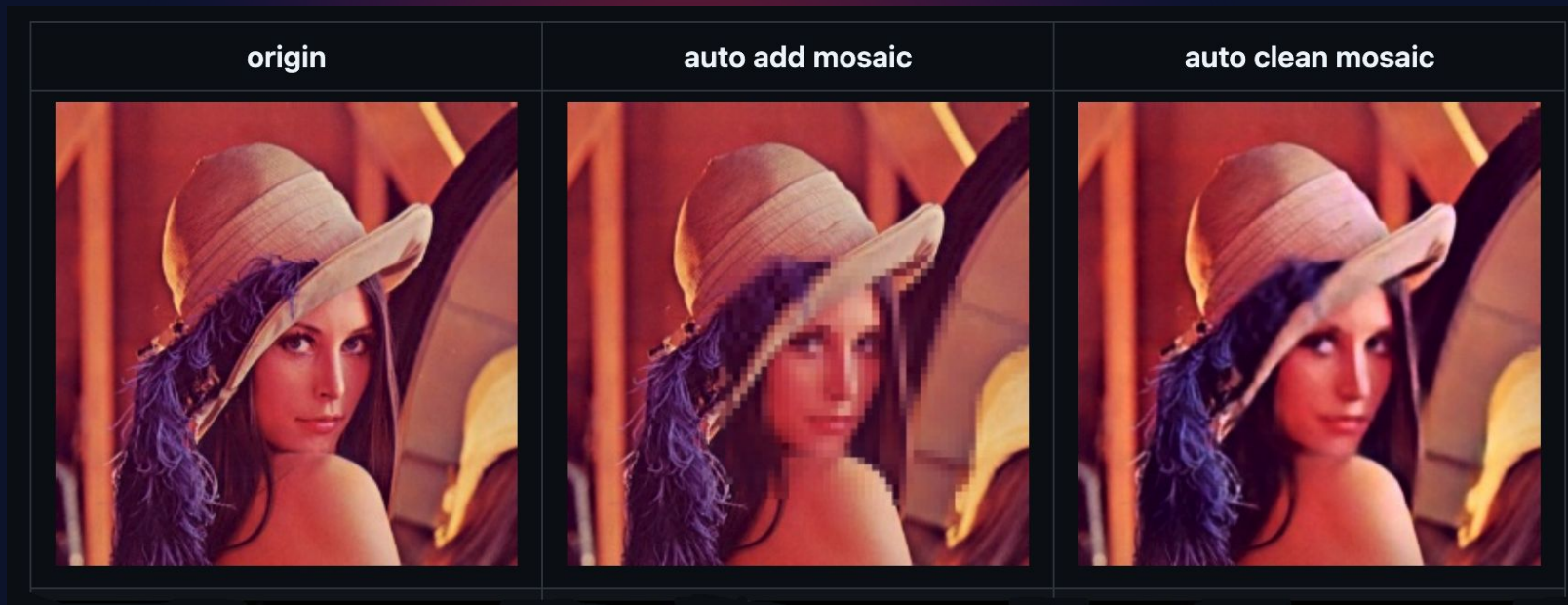
https://github.com/KoKuToru/de-pixelate_gaV-O6NPWrI

<https://cybernews.com/security/geerling-video-pixelation-challenge-cracked-in-hours/>

DeepMosaics: Restoration

<https://github.com/HypoX64/DeepMosaics>

- Works By Generative Adversarial Networks (GANs)



Demo: Depixelization (Grid Approx)

<https://youtu.be/DprDA-yliQA>

Demo: Unpixel (Generate → Test)

<https://youtu.be/aOwZOng8Ndo>

Demo: DeepMosaics (Restoration)

<https://youtu.be/KGJngYO45dA>

DeepMosaics: Video BEFORE



Image Attribution

https://github.com/KoKuToru/de-pixelate_gaV-O6NPWrI

<https://cybernews.com/security/geerling-video-pixelation-challenge-cracked-in-hours/>

DeepMosaics: Video AFTER

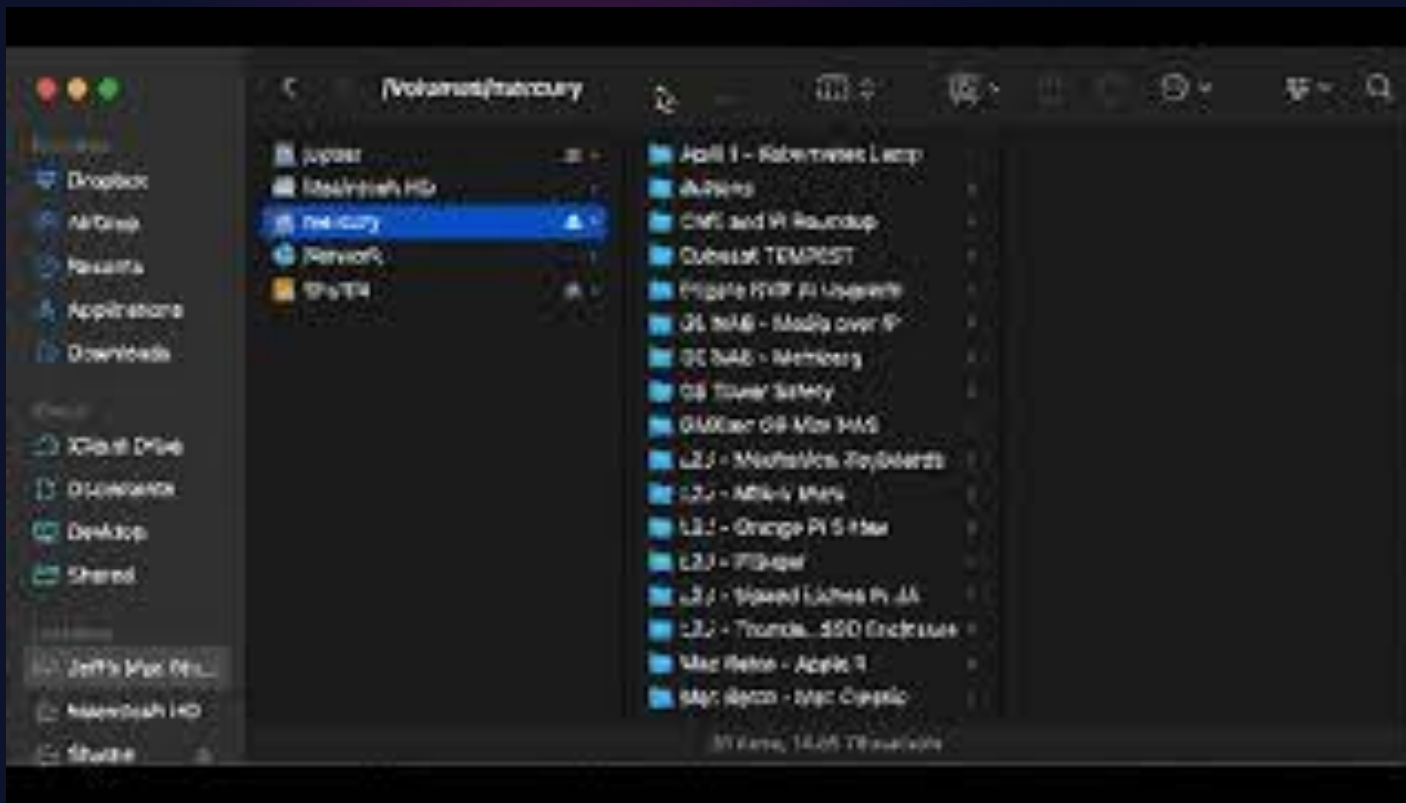


Image Attribution

https://github.com/KoKuToru/de-pixelate_gaV-O6NPWrl

<https://cybernews.com/security/geerling-video-pixelation-challenge-cracked-in-hours/>

Pushing the Envelope Using AI/ML

What Gaussian Blur and Text?

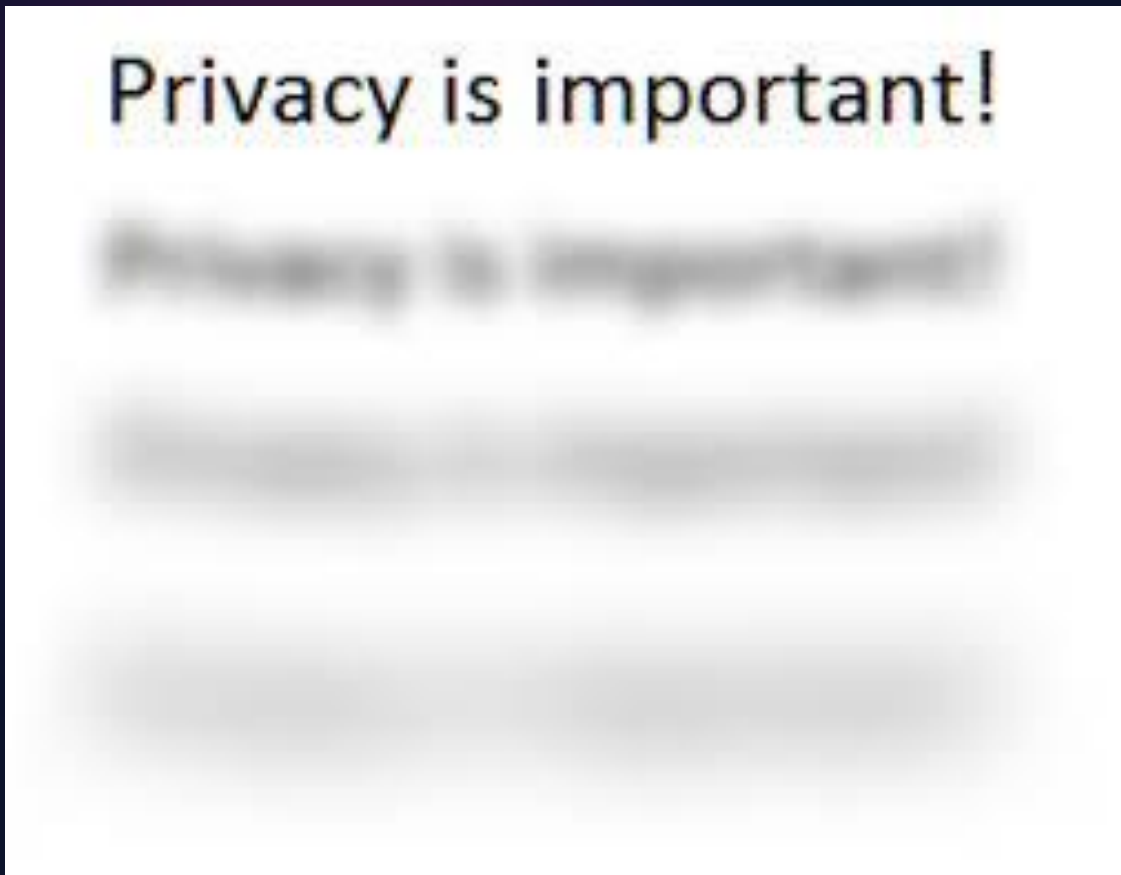


Image Attribution

https://www.researchgate.net/figure/Example-of-Gaussian-blurred-text-with-blur-radii-from-top-to-bottom-of-0p-15p-30p_fig5_305423573

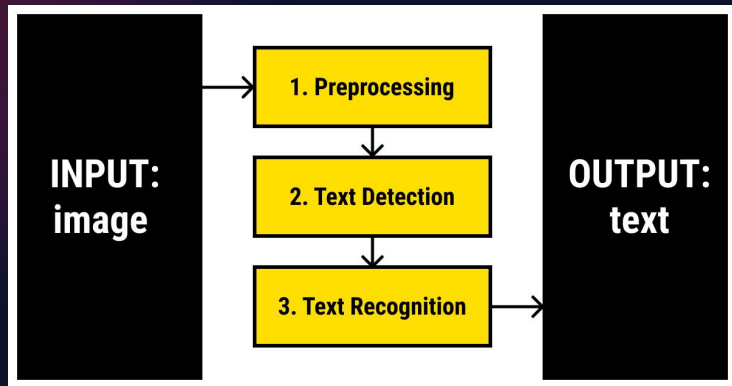
Skip Restoring Images

Most Forms Reversing Redaction Involves Restoration

- Depixelation: codeberg.org/spipm/Depixelization_poc
- Unpixel: github.com/oioio-space/unpixel
- DeepMosaics: github.com/HypoX64/DeepMosaics

Instead of Restoring, Let's Just Predict The Text/Data

- Predicts Character Sequence
- Users Exploiting The Service
- Legitimately Paid User



The Enemy From Within

What If Paid Customers, Use API to Create a Dataset?

- Just Make API Calls To Collect Data
- Build Representative Dataset
- Distillation Training

Sure, Against Terms of Service...

- That Won't Stop Anyone

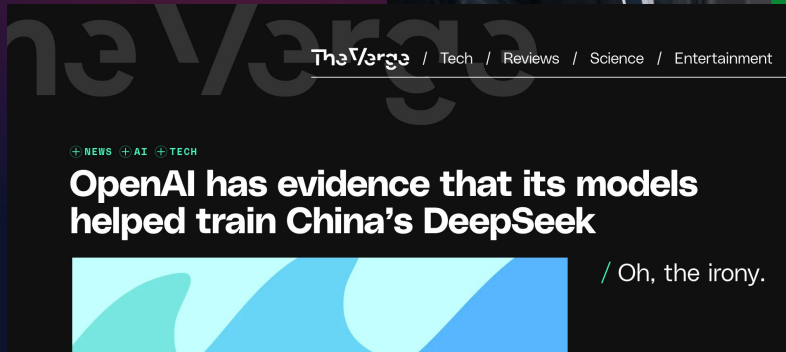


Image Attribution

<https://www.theverge.com/news/601195/openai-evidence-deepseek-distillation-ai-data>

<https://techcrunch.com/2026/04/30/elon-musk-testifies-that-xai-trained-grok-on-openai-models/>

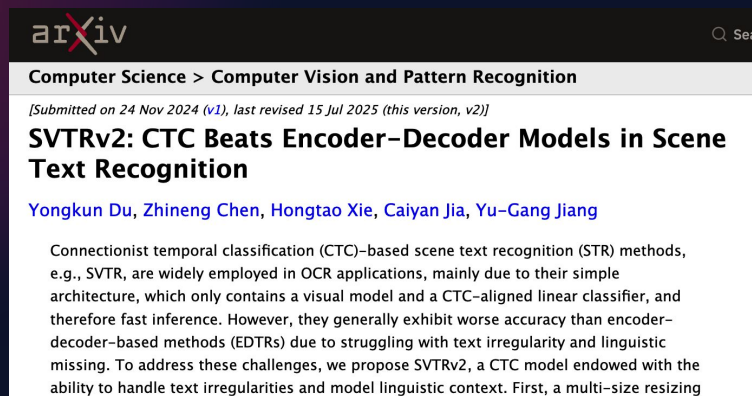
Model Architecture

SVTRv2 is a Connectionist Temporal Classification (CTC)-based Scene Text Recognition Model

- Map Unaligned Sequences For Tasks
 - Speech or Handwriting Recognition
- Outperforms Encoder-Decoder Methods

Incorporates:

- Multi-size Resizing Strategy
- Feature Rearrangement
- Trainable Semantic Guidance



Dataset Generation

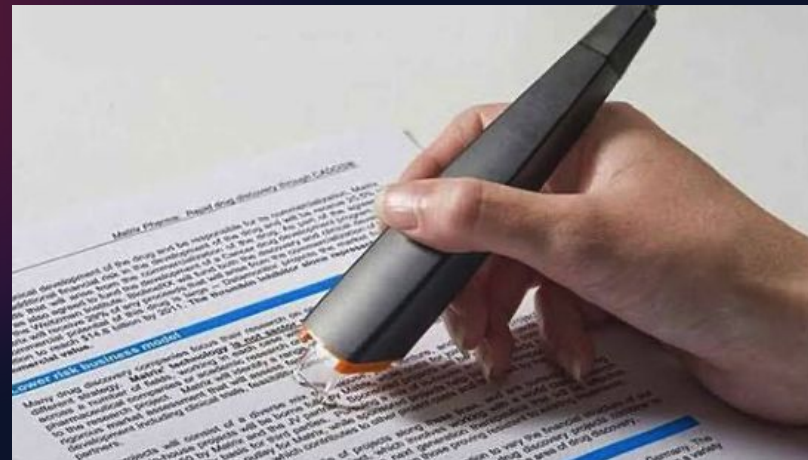
Dataset Composed Of:

- Training Input: Blurred Image (AFTER)
- Training Target: Original String

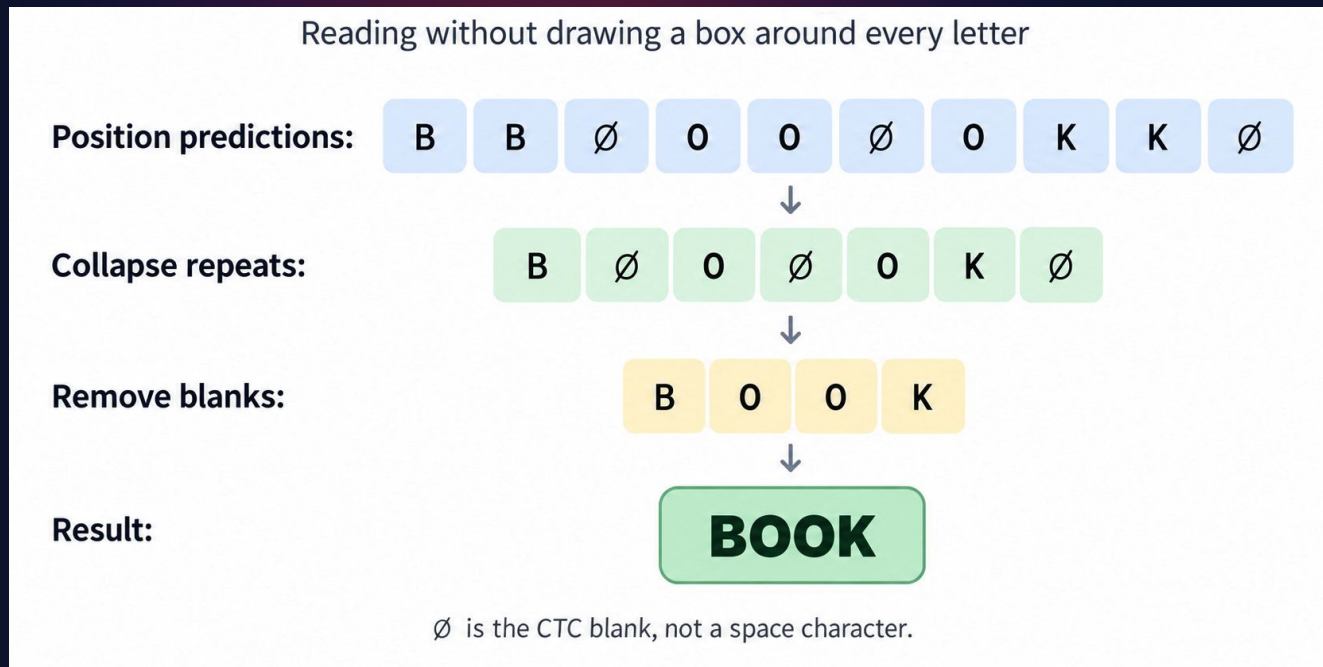
Clear Image (BEFORE) Not Needed

- Reference Material Only
- Not a Reconstruction Target

NOT Restoring Images!



What CTC Alignment Looks Like



Demo: Exploiting APIs & Gaussian Blur

<https://youtu.be/44pQGvDXAPc>

Why Specialize In One Method?

Specializing In a Single Problem Simplifies Things

- Blur Changes The Signal, Doesn't Remove It
- Recognizing Text Easier Than Reconstructing Pixels
- Consistent Pipeline Reduces Model Variability
- Learning Effort Can Be Reused
 - Doesn't Require Secret Access!
 - Reusable Across Similar Outputs!

Method/Principle Isn't Limited To Undoing Text Redaction

Resources

Resources

All Materials (Slides, Code, Instructions, etc):

github.com/davidvonthenen/2026-devox-xx-belgium

Devox Greece 2026:

[Sound of Your Secrets \(Acoustic Keystroke Logging\)](#)

The Secure Disclosure Podcast - Acoustic Keystroke Loggers:

- [Apple Podcast](#)
- [Spotify Podcast](#)
- [YouTube Podcast](#)

Disrupting AI and LLMs:

[WeAreDevelopers EU 2025: Confuse, Obfuscate, Disrupt](#)



Bonus Challenge!

github.com/2026-devoxx-belgium/demos/bonus/surprise-bonus.pdf

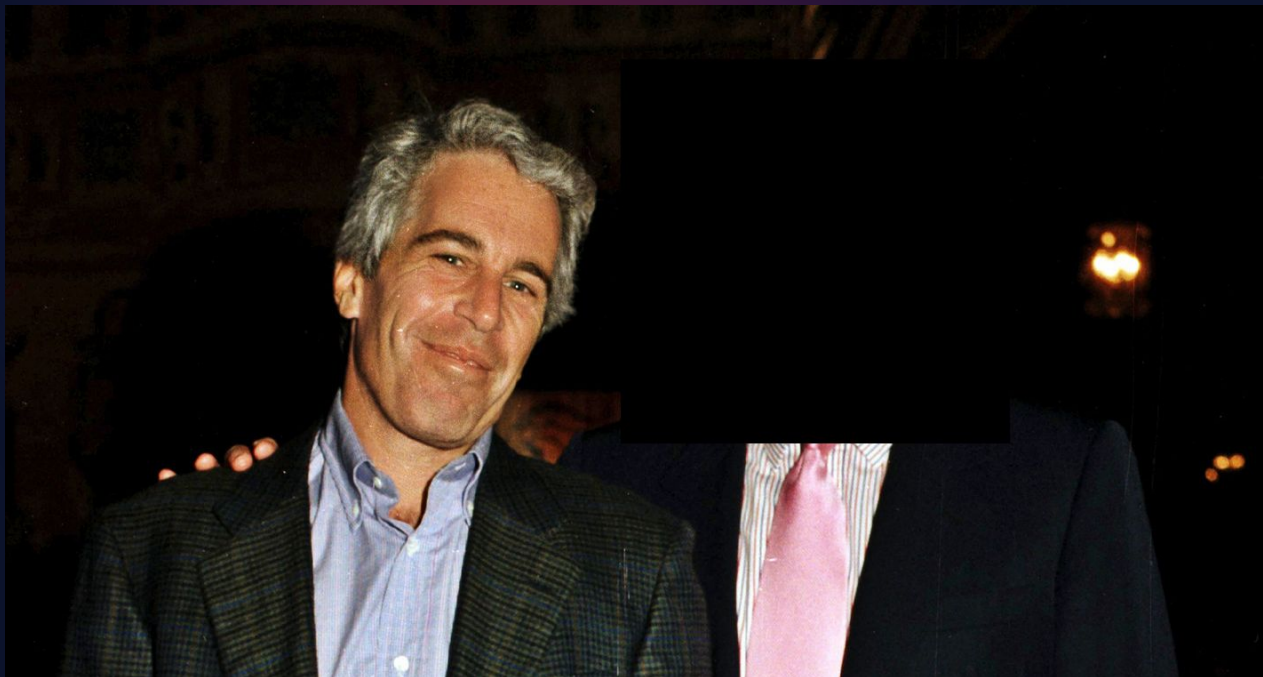


Image Attribution

<https://www.pbs.org/newshour/politics/the-facts-and-timeline-of-trump-and-epsteins-falling-out>

Thank You!



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